Task 2: Client Profitability and Spread Recommendation

Classification of Clients

Clients are classified as *net-profitable* or *net-costly* based on their Expected Aggregate PnL per trade (using uniform closing weights across all horizons $\tau \in \{5, 10, 15, 20, 25, 30\}$).

Based on the quantitative results derived from the historical trade data:

- **Profitable Clients (A, B, C, D):** These clients yield a positive Aggregate PnL. Clients A (1.90), B (1.63), and C (1.17) provide strong, consistent profitability. Client D yields a marginally positive Aggregate PnL (0.30) but remains mathematically net-profitable under the historical execution conditions.
- **Costly Clients (E, F):** These clients possess a negative Aggregate PnL. Client E results in an expected loss of -0.40 per trade, while Client F is highly toxic, costing the Liquidity Provider (LP) -1.44 per trade.

Spread Recommendation and Adversity Profile Relation

The minimum half-spread, δ^*_{client}, defines the buffer required to make the Aggregate PnL non-negative if we hypothetically quoted every trade exactly at $M_0 \pm \delta^*_{client}$. Conceptually, δ^*_{client} represents the volume-weighted expected adverse mid-price drift ($M_{avg} - M_0$).

The computed minimum half-spreads heavily corroborate the adversity profiles observed in Task 1:

- **Uninformed Flow (Clients A, B, C):** $\delta^* = 0.0$. For these clients, the required spread to break even against the M_0 drift is zero (mathematically negative, but conceptually floored to 0.0). This perfectly aligns with their Task 1 adversity profiles, which remained safely below 50% across all horizons. Because their flow is benign and does not systematically predict price movements, the LP does not need a protective spread to offset adverse selection.
- **Marginally Toxic Flow (Client D):** $\delta^* \approx 0.0096$. While Client D was technically profitable historically, quoting them precisely at mid requires a small positive spread just to break even. This bridges the gap between Task 1 and Task 2: in Task 1, Client D's adversity crossed the 50% threshold at longer horizons ($\tau \geq 25$). The positive δ^* directly quantifies the spread needed to absorb this latent, medium-term adverse price drift.
- **Informed Flow (Clients E, F):** $\delta^* \approx 0.019$ and 0.033, respectively. These clients mandate significant half-spreads to achieve breakeven profitability. This is the direct monetary consequence of their toxic adversity profiles from Task 1 (where Client F reached nearly 62% adversity). Because informed clients consistently trade in the direction of imminent price momentum, the LP is systematically left on the wrong side of the drift (M_{avg} moves against the LP's position). A high δ^*_{client} is mathematically necessary to pre-emptively offset the high probability of a negative outcome.